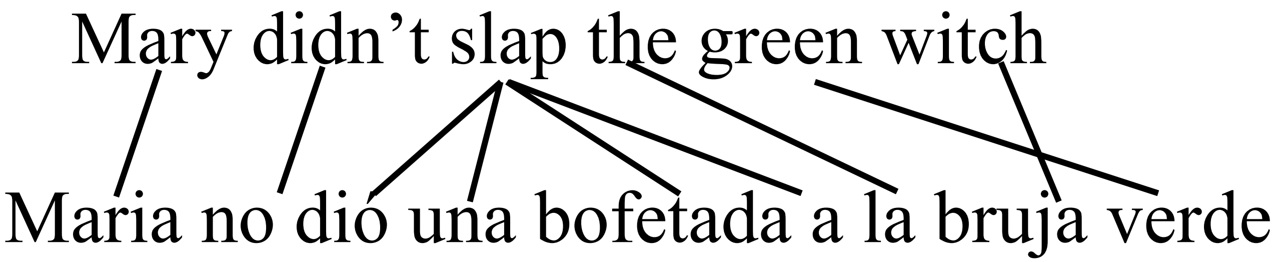
Statistical Methods

- As with other areas of MT, manually crafted rules are low in coverage and don't scale well
- SMT acquires knowledge needed for translation from a ***parallel corpus*** or ***bitext*** that contains the same set of documents in two languages
  - With some models more supervision is required, such as a syntactic parser or a bi-lingual dictionary
- The *Canadian Hansards* (parliamentary proceedings in French and English) and the *Europarl* (protocols of the European Parliament, in all official EU languages) are well-known parallel corpora
  - Parallel corpora are also published through the annual conference for MT (WMT)

Word Alignment

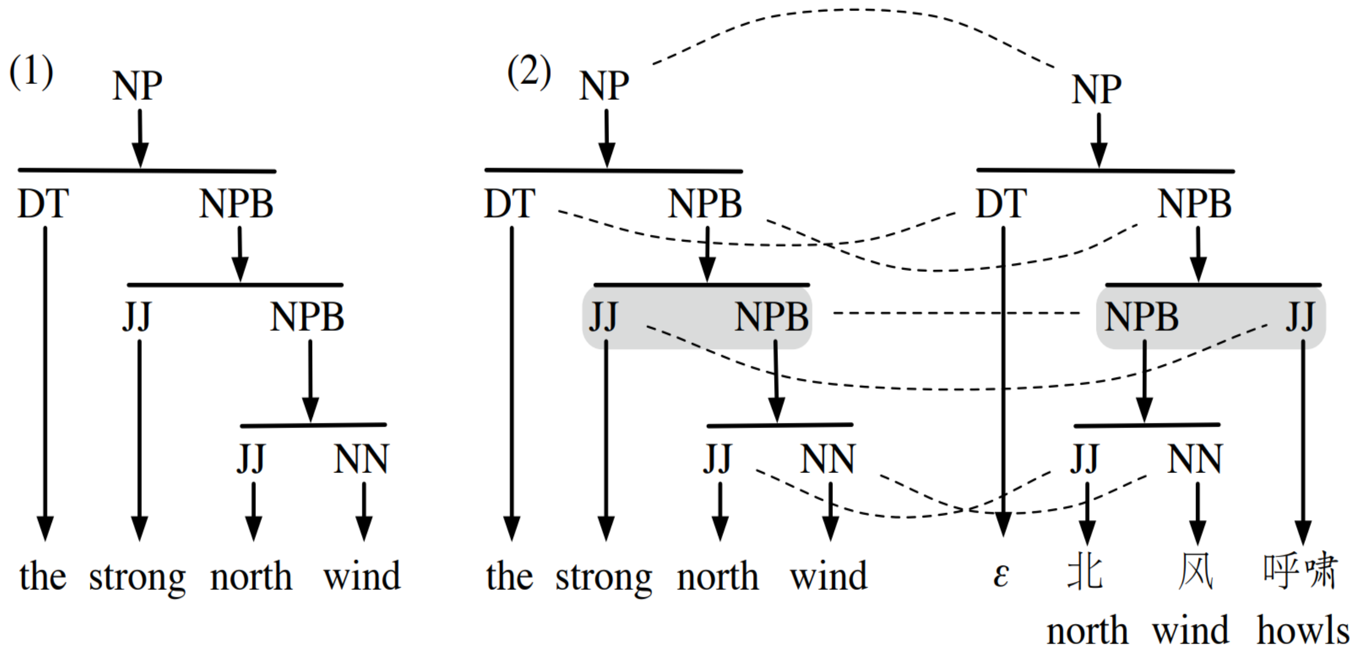
- One of the basic tools in MT
- Shows mapping between words in one language and the other



- Often done by finding co-occurrences across of words/phrases across parallel data

Syntax-Based Statistical Machine Translation

- Syntax-based approaches demonstrated improvement for translating between more distant language pairs, e.g., Chinese/English
- They assume that the syntactic trees of the derivations in the two languages are similar, except that each right-hand-side of a production may be permuted relative to the production in the other language
- These transformations are learned from syntax-annotated parallel data



Adapted from Lopez (2007) Survey on SMT

Syntax-Based Statistical Machine Translation

- Despite the promise of such models, their assumption is still too weak to deal with all cross-linguistic divergences where the syntactic tree is entirely different

Jacky happened to meet the guy yesterday

ג'קי פגש במקרה את הבחור אתמול

ג'קי קרה לפגוש את הבחור אתמול